

ORIGINAL RESEARCH ARTICLE — SYSTEMATIC REVIEW AND META-ANALYSIS

In adults receiving CAR-T cell therapy, does early administration of IL-6 inhibitors (such as tocilizumab) reduce the incidence of severe (Grade 3 or higher) cytokine release syndrome compared with standard or delayed treatment?

Protocol & Run Identifier: run-20260921-073015 | Timestamp (UTC): 2026-09-21T07:30:15 | Execution Duration: 150.8s | Orchestrator: gemini-3.6-flash | Clinical Reviewer: ollama_chat/medgemma

STRUCTURED ABSTRACT

Background & Objective: To systematically evaluate clinical evidence addressing: *In adults receiving CAR-T cell therapy, does early administration of IL-6 inhibitors (such as tocilizumab) reduce the incidence of severe (Grade 3 or higher) cytokine release syndrome compared with standard or delayed treatment?*. Specifically, this review assesses **Early administration of IL-6 inhibitors (such as tocilizumab)** compared with **Standard or delayed treatment** with respect to **Incidence of severe (Grade 3 or higher) cytokine release syndrome** in **Adults receiving CAR-T cell therapy**.

Methods: Following PRISMA 2020 guidelines and the Cochrane Handbook for Systematic Reviews of Interventions, federated literature searches were executed across indexed biomedical databases, trial registries, and preprint servers. Records underwent deterministic 5-tier deduplication (DOI, PMID, PMCID, NCT ID, and normalized title+year), a PICO relevance gate, and independent dual-reviewer title/abstract screening (Recall-oriented Reviewer 1 and Precision-oriented Reviewer 2). Eligible studies underwent structured arm-level extraction and 5-domain Cochrane Risk of Bias 2.0 (RoB 2) assessment. Certainty of evidence was rated using the GRADE framework.

Results: Of **531** records identified across databases, **28** duplicates were removed, leaving **503** unique records. Out of **20** records screened, **20** were excluded and **0** met eligibility criteria. Data extraction and Cochrane RoB 2 assessment were completed for **0** studies.

Conclusion & Synthesis Status: Fewer than 2 extracted abstracts reported complete two-arm comparative event counts or means/SDs; per Cochrane Handbook guardrails, findings are synthesized qualitatively across all 0 included studies in Table 1 without imputing unreported comparator data.

1. Protocol Specification (PICOS) & Literature Search Strategy

PICOS Parameter	Protocol Definition & Eligibility Rules
Population (P)	Adults receiving CAR-T cell therapy
Intervention (I)	Early administration of IL-6 inhibitors (such as tocilizumab)
Comparator (C)	Standard or delayed treatment
Primary Outcome (O)	Incidence of severe (Grade 3 or higher) cytokine release syndrome
Eligible Study Designs (S)	Randomized Controlled Trial
Inclusion Criteria	Adults receiving CAR-T cell therapy; Intervention group: Early administration of IL-6 inhibitors (such as tocilizumab); Comparator group: Standard or delayed treatment; Severe (Grade 3 or higher) cytokine release syndrome as the primary outcome
Exclusion Criteria	Animal studies; In-vitro studies; Narrative reviews; Editorials; Case reports

Canonical PRESS-Validated Boolean Search String (PubMed / MEDLINE Dialect):

("CAR-T cell therapy"[tiab] OR "CAR T-cell Therapy"[tiab] OR CAR-T[tiab] OR "CAR T"[tiab] OR "CAR T-cell Therapy"[MeSH Terms] OR "CAR-T cell Therapy"[MeSH Terms] OR "CAR-T"[MeSH Terms] OR "CAR T"[MeSH Terms]) AND ("IL-6 inhibitors"[tiab] OR IL-6[tiab] OR tocilizumab[tiab] OR "IL-6 blockade"[tiab] OR "IL-6 inhibition"[tiab] OR "Interleukin-6"[MeSH Terms])

2. PRISMA 2020 Study Selection Flow & Complete Record Accounting

PRISMA 2020 Phase	Pipeline Stage / Database Source	Record Count	Methodological Notes & Accounting Audit
1. Identification	Database: clinicaltrials	100	API Status: success
1. Identification	Database: europepmc	100	API Status: success

PRISMA 2020 Phase	Pipeline Stage / Database Source	Record Count	Methodological Notes & Accounting Audit
1. Identification	Database: pubmed	100	API Status: success
1. Identification	Database: crossref	100	API Status: success
1. Identification	Database: openalex	100	API Status: success
1. Identification	Database: semantic_scholar	0	API Status: success
1. Identification	Database: preprints	31	API Status: success
1. Identification	Total Records Identified	531	Combined citations retrieved across all active connectors
2. Deduplication	Duplicate Records Removed	-28	5-tier deterministic match (DOI / PMID / NCT / Title+Year)
2. Deduplication	Unique Records After Deduplication	503	531 identified - 28 duplicates = 503 unique records
3. Screening	Unscreened (Beyond Screening Cap)	-483	Screening cap configured at 20 records
3. Screening	Records Screened (Title & Abstract)	20	Independent dual MedGemma reviewers (Inter-reviewer agreement: N/A)
3. Screening	Records Excluded at Title/Abstract Screening	-20	20 by deterministic PICO gate + 0 by dual reviewers (Table 2)
3. Screening	Studies Approved at Screening	0	20 screened - 20 excluded = 0 eligible studies
4. Eligibility / Extraction	Studies Extracted & Assessed for RoB 2	0	Structured arm-level extraction + 5-domain Cochrane RoB 2.0
5. Inclusion	Studies Included in Quantitative Meta-Analysis	0	0 total participants analyzed quantitatively

3. Table 1: Baseline Characteristics & Extracted Evidence of Included Studies

Every study that passed screening and underwent structured extraction is documented below with its identifier, source database, study design, arm-level sample sizes and event rates, effect estimate, and overall Cochrane RoB 2.0 judgement:

4. Methodological Quality & Cochrane Risk of Bias 2.0 (RoB 2) Assessment

Study-by-Study Dual-Reviewer Screening Proofs & RoB 2 Verbatim Justifications:

5. Quantitative Meta-Analysis, Estimator Sensitivity & Visual Figures

Quantitative two-arm pooling was not executed because fewer than 2 included abstracts provided complete two-arm comparative event counts or means/SDs. Per Cochrane Handbook standards, unreported comparator figures are never imputed.

Reproducible R Verification Script (meta / metafor):

```
# =====
# Reproducible R Meta-Analysis Script – Run ID: run-20260921-073015
# Outcome: Incidence of severe (Grade 3 or higher) cytokine release syndrome
# Generated per PRISMA 2020 & Cochrane Handbook specifications
# =====
library(meta)
library(metafor)

# Fewer than 2 studies reported comparative arm-level event counts or means/SDs
# in the retrieved abstracts. Populate m_data below after full-text extraction:
m_data <- data.frame(
  study = c("Study_1", "Study_2"),
  event_e = c(NA, NA), n_e = c(NA, NA),
  event_c = c(NA, NA), n_c = c(NA, NA)
)
```

7. Transparency Audit Appendix: Excluded, Unpooled & Duplicate Records

Table 2: Articles Excluded at Title/Abstract Screening (n = 20)

# / Article ID	Title	PRISMA Category	Decision Source	Documented Exclusion Reason / Proof
1. NCT04499573	Bispecific CD19/CD22 CAR-T for Treatment of Children and Young Adults With r/r B-ALL	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (IL-6 inhibitors, IL-6, tocilizumab).

# / Article ID	Title	PRISMA Category	Decision Source	Documented Exclusion Reason / Proof
2. NCT06925022	Educational Programs Based on Healthy Habits to Improve Quality of Life and Psychosocial Profile in Women With Neurodegenerativ...	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the population (CAR-T cell therapy, CAR T-cell Therapy, CAR-T) or the intervention (IL-6 inhibitors, IL-6, tocilizumab).
3. NCT06716164	Safety and Efficacy of Metabolically Armed CD19 CAR-T Cells (Meta10-19) in the Treatment of r/r B-NHL Clinical Research	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (IL-6 inhibitors, IL-6, tocilizumab).
4. NCT07033299	The Safety, Efficacy, and Cellular Metabolic Kinetics of CT1192 in Treating Patients With Anti Neutrophil Cytoplasmic Antibody...	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (IL-6 inhibitors, IL-6, tocilizumab).
5. NCT06228404	Clinical Study of Safety and Efficacy of Enhanced PSMA CAR- T in Refractory CRPC	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (IL-6 inhibitors, IL-6, tocilizumab).
6. NCT06106893	A Clinical Study of CD19 Universal CAR-γδT Cells in Active Systemic Lupus Erythematosus	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the population (CAR-T cell therapy, CAR T-cell Therapy, CAR-T) or the intervention (IL-6 inhibitors, IL-6, tocilizumab).
7. NCT07456371	PIC1 Injection Therapy for Relapsed/Refractory B-NHL	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (IL-6 inhibitors, IL-6, tocilizumab).
8. NCT06407947	Study of CT071 Injection in High Risk Newly Diagnosed Multiple Myeloma	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the population (CAR-T cell therapy, CAR T-cell Therapy, CAR-T) or the intervention (IL-6 inhibitors, IL-6, tocilizumab).
9. 10.1016/j.jtct.2026.05.045	Clinical Study of SenL-T7 CAR T Cells in the Treatment of Relapsed and Refractory CD7+ T-cell Lymphoblastic Leukemia or T-cell...	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (IL-6 inhibitors, IL-6, tocilizumab).
10. NCT07563543	RN1701 Injection in the Treatment of Relapsed/Refractory B-Cell Lymphomas	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (IL-6 inhibitors, IL-6, tocilizumab).
11. NCT03146533	CD19 CART Cells for Patients With Relapse and Refractory CD19+ B-cell Lymphoma.	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the population (CAR-T cell therapy, CAR T-cell Therapy, CAR-T) or the intervention (IL-6 inhibitors, IL-6, tocilizumab).
12. NCT05473221	Evaluate the Safety and Efficacy of CD33 CAR-T in Patients With R/R AML	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (IL-6 inhibitors, IL-6, tocilizumab).
13. NCT06987916	Efficacy and Safety Evaluation of U01(ssCART-19) in B-Cell Lymphoma	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the population (CAR-T cell therapy, CAR T-cell Therapy, CAR-T).
14. NCT05667506	A Study of CNCT19 Treatment in Children and Adolescent r/r ALL Patients(Pediatric)	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (IL-6 inhibitors, IL-6, tocilizumab).
15. NCT03633773	Safety and Efficacy Evaluation of MUC-1 CART in the Treatment of Intrahepatic Cholangiocarcinoma	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (IL-6 inhibitors, IL-6, tocilizumab).
16. 10.1016/j.clim.2022.109030	A Study to Evaluate the Safety and Efficacy of A2B395, an Allogeneic Logic-gated CAR T, in Participants With Solid Tumors That...	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (IL-6 inhibitors, IL-6, tocilizumab).
17. NCT04546906	Safety and Efficacy Study of CD22 CAR-T Cells for Relapsed or Refractory Acute Lymphoblastic Leukemia	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (IL-6 inhibitors, IL-6, tocilizumab).
18. NCT06612645	Safety and Efficacy of CMD03 CAR T Cell in Children With Relapse or Refractory Solid Tumors	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (IL-6 inhibitors, IL-6, tocilizumab).

# / Article ID	Title	PRISMA Category	Decision Source	Documented Exclusion Reason / Proof
19. NCT07078929	Clinical Trial of CMD63 Chimeric Antigen Receptor T-cell (CAR T-cell) in Children With Acute Lymphoblastic Leukemia (ALL)	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (IL-6 inhibitors, IL-6, tocilizumab).
20. NCT06481735	TCR Reserved and Power3 (SPPL3) Gene Knock-out Allogeneic CD19-targeting CAR-T Cell Therapy in r/r B-ALL	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (IL-6 inhibitors, IL-6, tocilizumab).

Table 4: Deduplication Audit Log — Removed Duplicate Records (n = 28)

# / Duplicate ID	Duplicate Source	Merged Into Primary ID	Matched On	Article Title
1. NCT04499573	europemc	NCT04499573	nct_id	Bispecific CD19/CD22 CAR-T for Treatment of Children and Young Adults With r/r B-ALL
2. NCT06925022	pubmed	NCT06925022	nct_id	Educational Programs Based on Healthy Habits to Improve Quality of Life and Psychosocial Profile in Women With Neurodegenerativ...
3. NCT06716164	crossref	NCT06716164	nct_id	Safety and Efficacy of Metabolically Armed CD19 CAR-T Cells (Meta10-19) in the Treatment of r/r B-NHL Clinical Research
4. NCT07033299	openalex	NCT07033299	nct_id	The Safety, Efficacy, and Cellular Metabolic Kinetics of CT1192 in Treating Patients With Anti Neutrophil Cytoplasmic Antibody...
5. NCT06228404	semantic_scholar	NCT06228404	nct_id	Clinical Study of Safety and Efficacy of Enhanced PSMA CAR-T in Refractory CRPC
6. NCT06106893	preprints	NCT06106893	nct_id	A Clinical Study of CD19 Universal CAR-γδT Cells in Active Systemic Lupus Erythematosus
7. NCT07456371	clinicaltrials	NCT07456371	nct_id	PIC1 Injection Therapy for Relapsed/Refractory B-NHL
8. NCT06407947	europemc	NCT06407947	nct_id	Study of CT071 Injection in High Risk Newly Diagnosed Multiple Myeloma
9. 10.1016/j.jtct.2026.05.045	pubmed	10.1016/j.jtct.2026.05.045	doi	Clinical Study of SenL-T7 CAR T Cells in the Treatment of Relapsed and Refractory CD7+ T-cell Lymphoblastic Leukemia or T-cell...
10. NCT07563543	crossref	NCT07563543	nct_id	RN1701 Injection in the Treatment of Relapsed/Refractory B-Cell Lymphomas
11. NCT03146533	openalex	NCT03146533	nct_id	CD19 CART Cells for Patients With Relapse and Refractory CD19+ B-cell Lymphoma.
12. NCT05473221	semantic_scholar	NCT05473221	nct_id	Evaluate the Safety and Efficacy of CD33 CAR-T in Patients With R/R AML
13. NCT06987916	preprints	NCT06987916	nct_id	Efficacy and Safety Evaluation of U01(ssCART-19) in B-Cell Lymphoma
14. NCT05667506	clinicaltrials	NCT05667506	nct_id	A Study of CNCT19 Treatment in Children and Adolescent r/r ALL Patients(Pediatric)
15. NCT03633773	europemc	NCT03633773	nct_id	Safety and Efficacy Evaluation of MUC-1 CART in the Treatment of Intrahepatic Cholangiocarcinoma
16. 10.1016/j.clim.2022.109030	pubmed	10.1016/j.clim.2022.109030	doi	A Study to Evaluate the Safety and Efficacy of A2B395, an Allogeneic Logic-gated CAR T, in Participants With Solid Tumors That...
17. NCT04546906	crossref	NCT04546906	nct_id	Safety and Efficacy Study of CD22 CAR-T Cells for Relapsed or Refractory Acute Lymphoblastic Leukemia
18. NCT06612645	openalex	NCT06612645	nct_id	Safety and Efficacy of CMD03 CAR T Cell in Children With Relapse or Refractory Solid Tumors
19. NCT07078929	semantic_scholar	NCT07078929	nct_id	Clinical Trial of CMD63 Chimeric Antigen Receptor T-cell (CAR T-cell) in Children With Acute Lymphoblastic Leukemia (ALL)
20. NCT06481735	preprints	NCT06481735	nct_id	TCR Reserved and Power3 (SPPL3) Gene Knock-out Allogeneic CD19-targeting CAR-T Cell Therapy in r/r B-ALL
21. NCT04499573	clinicaltrials	NCT04499573	nct_id	Bispecific CD19/CD22 CAR-T for Treatment of Children and Young Adults With r/r B-ALL

# / Duplicate ID	Duplicate Source	Merged Into Primary ID	Matched On	Article Title
22. NCT06925022	europepmc	NCT06925022	nct_id	Educational Programs Based on Healthy Habits to Improve Quality of Life and Psychosocial Profile in Women With Neurodegenerativ...
23. NCT06716164	pubmed	NCT06716164	nct_id	Safety and Efficacy of Metabolically Armed CD19 CAR-T Cells (Meta10-19) in the Treatment of r/r B-NHL Clinical Research
24. NCT07033299	crossref	NCT07033299	nct_id	The Safety, Efficacy, and Cellular Metabolic Kinetics of CT1192 in Treating Patients With Anti Neutrophil Cytoplasmic Antibody...
25. NCT06228404	openalex	NCT06228404	nct_id	Clinical Study of Safety and Efficacy of Enhanced PSMA CAR-T in Refractory CRPC
26. NCT06106893	semantic_scholar	NCT06106893	nct_id	A Clinical Study of CD19 Universal CAR-γδT Cells in Active Systemic Lupus Erythematosus
27. NCT07456371	preprints	NCT07456371	nct_id	PIC1 Injection Therapy for Relapsed/Refractory B-NHL
28. NCT06407947	clinicaltrials	NCT06407947	nct_id	Study of CT071 Injection in High Risk Newly Diagnosed Multiple Myeloma